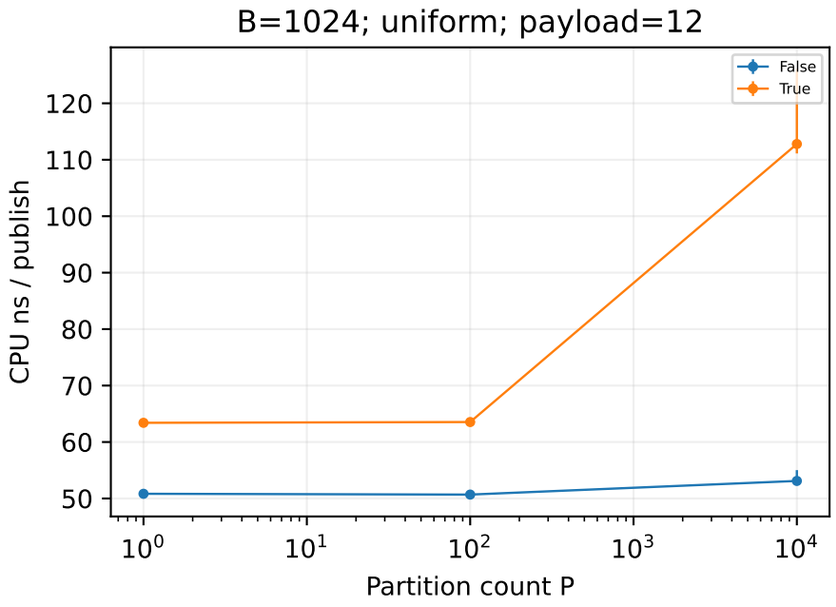
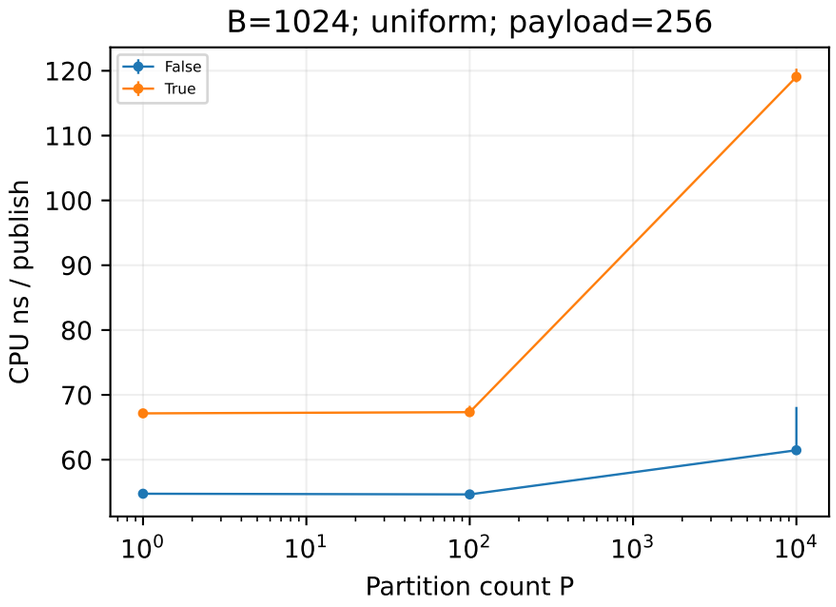
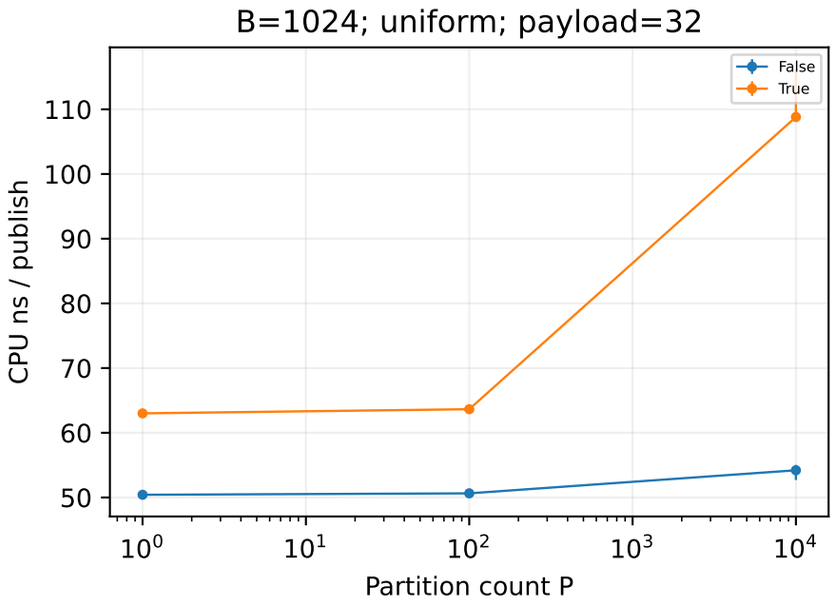
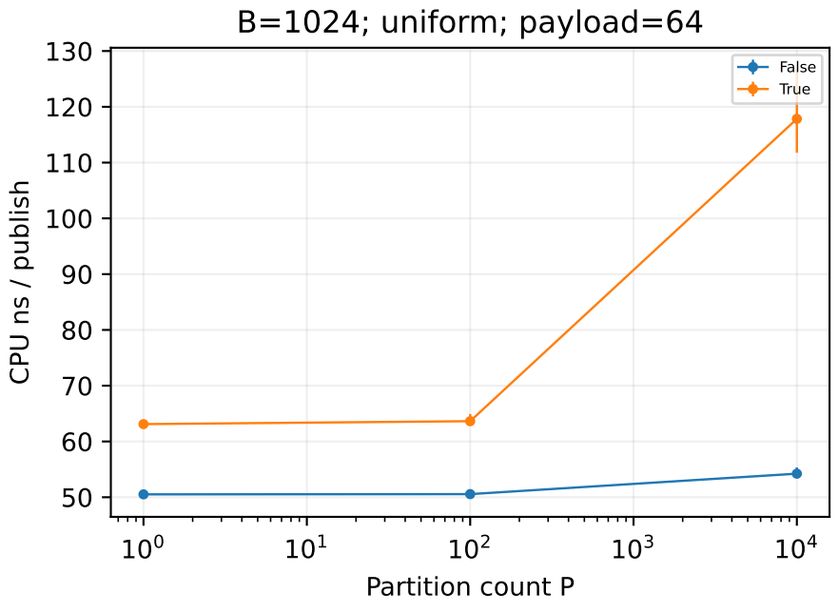
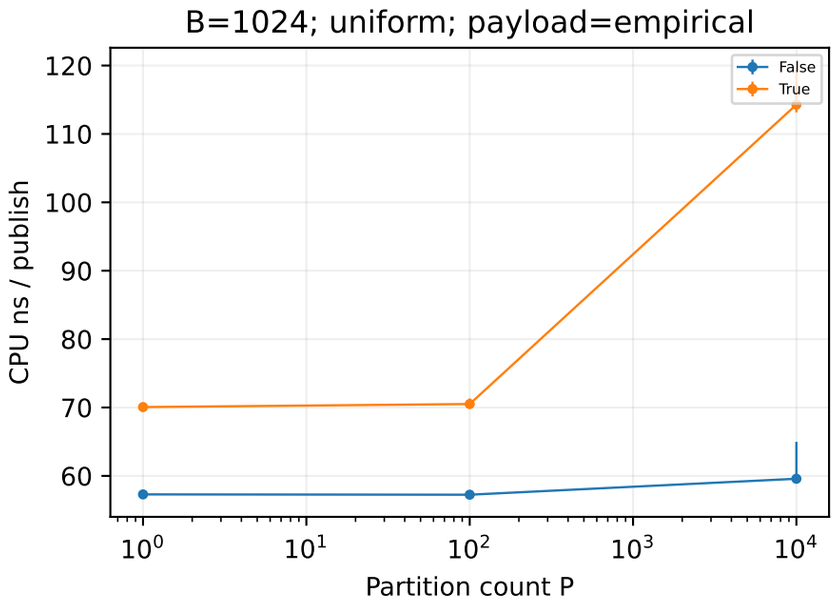
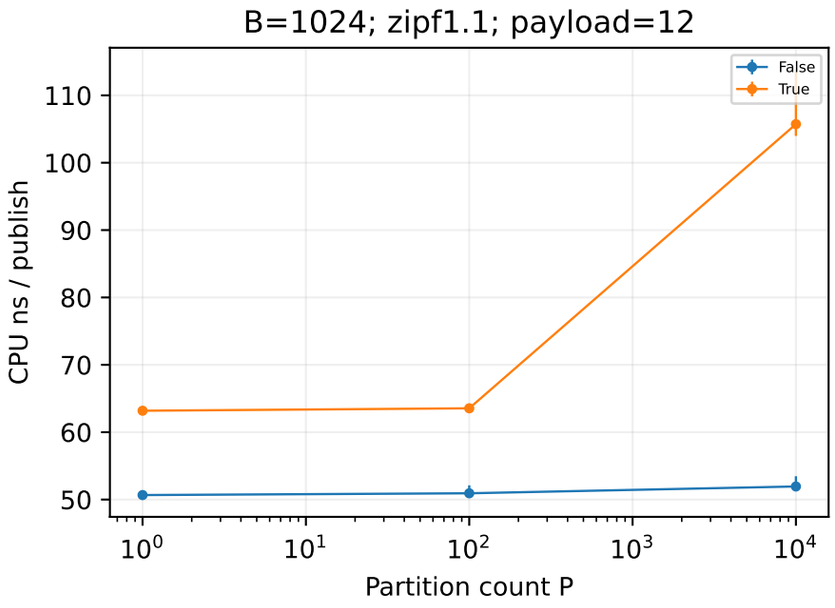


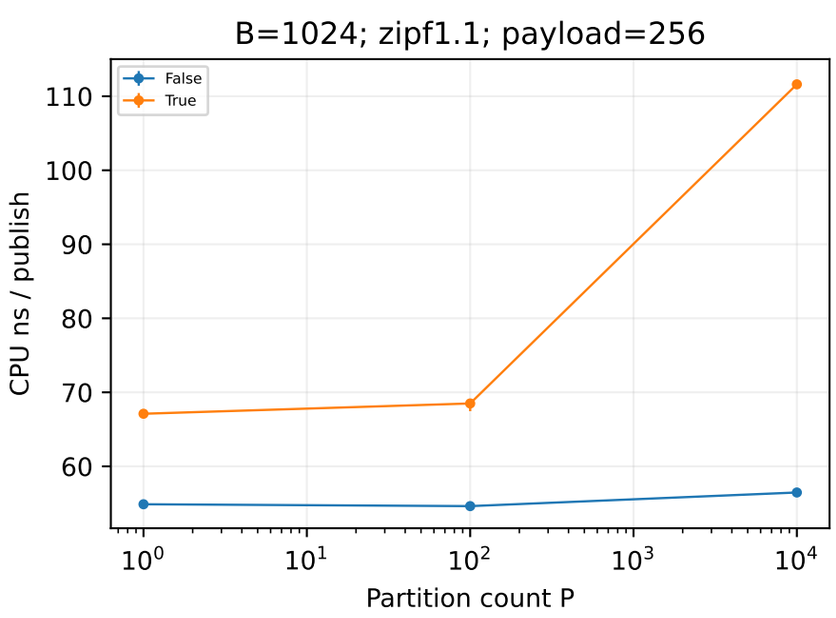


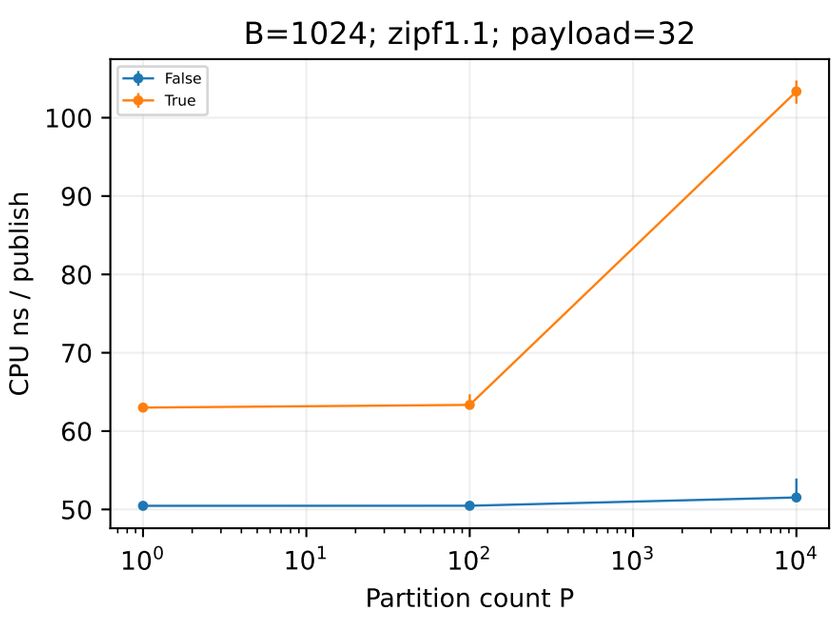


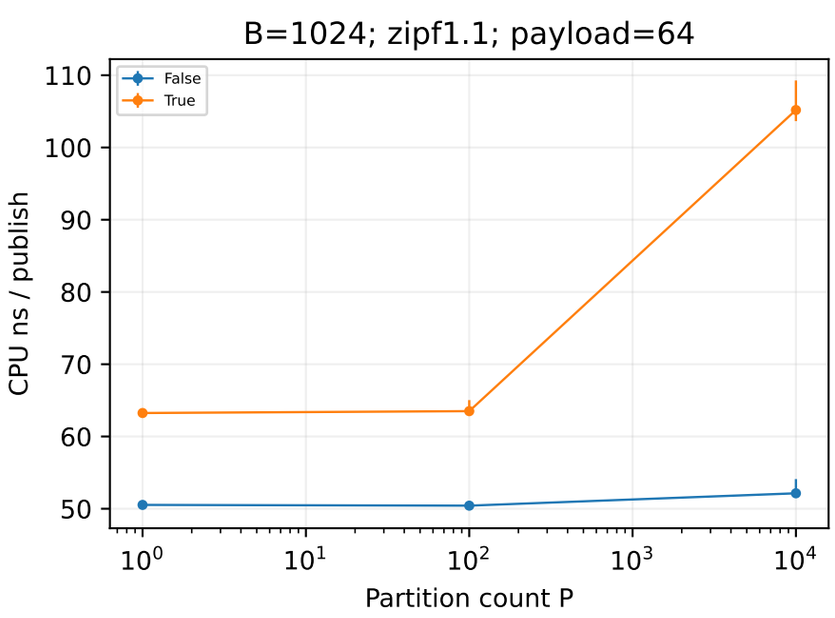


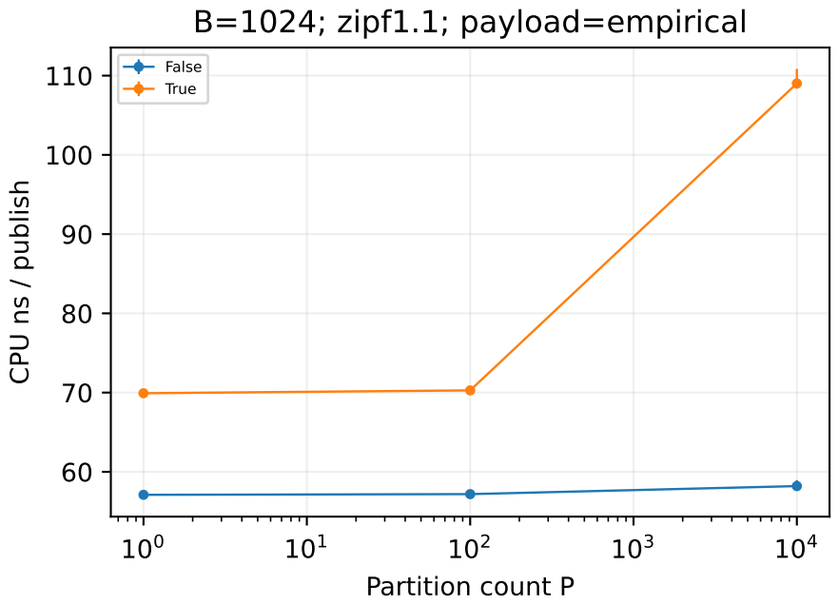


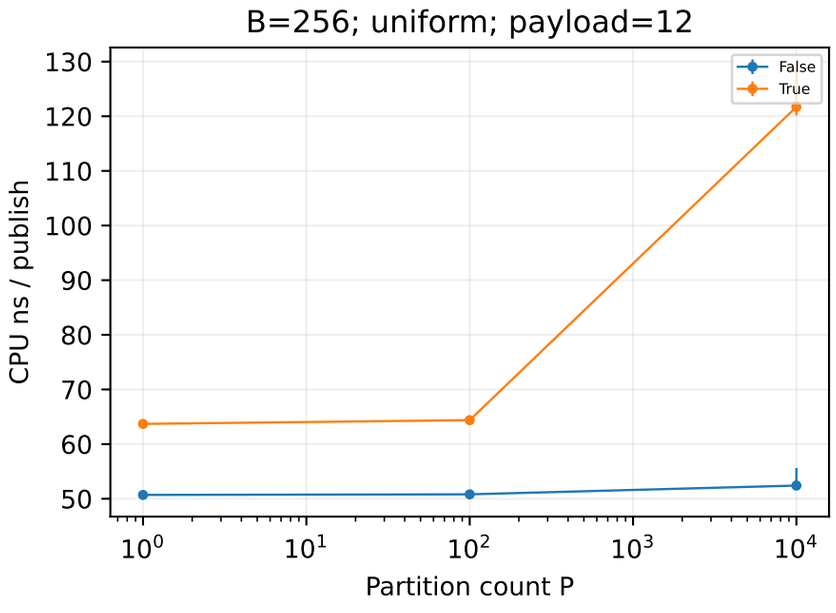


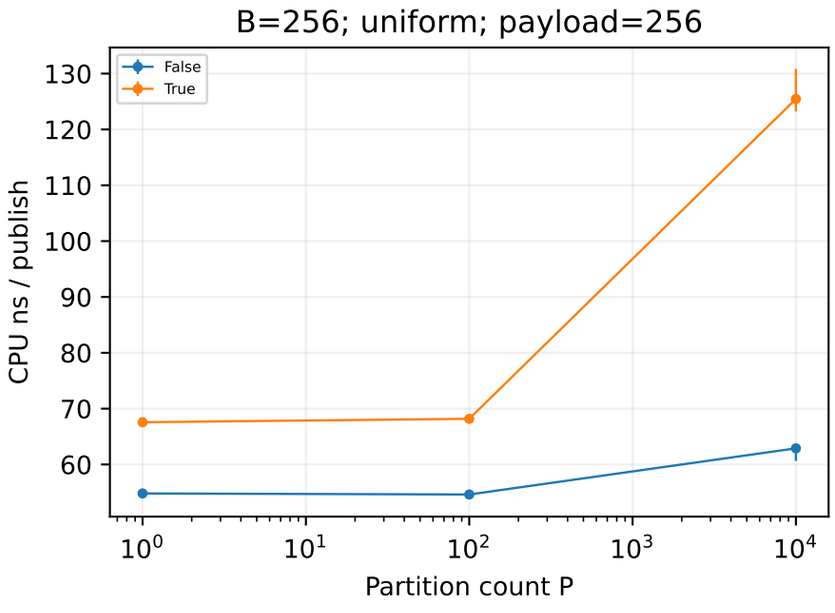


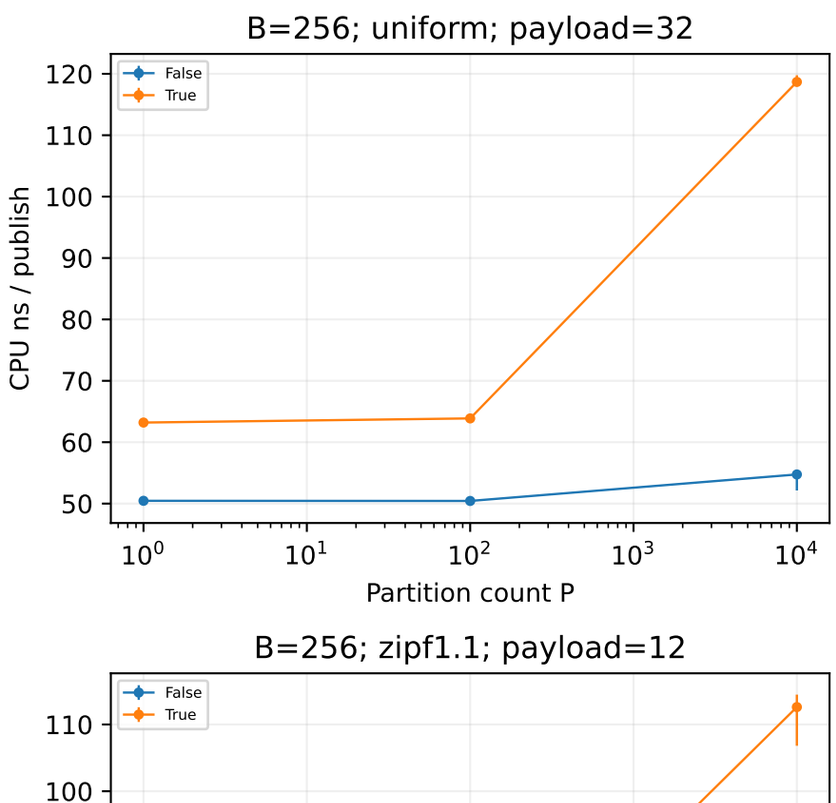




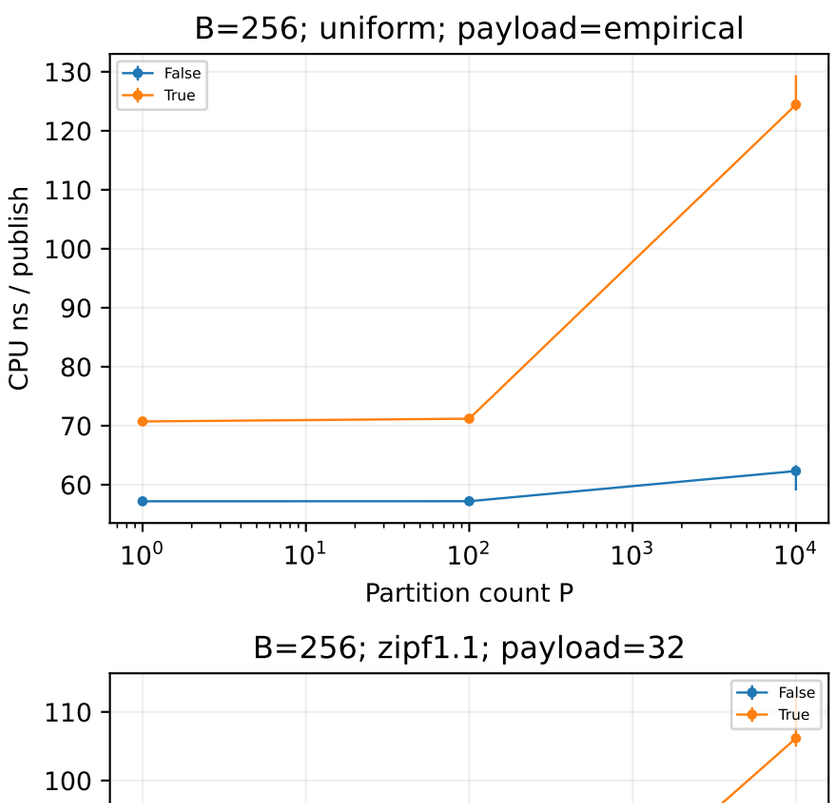


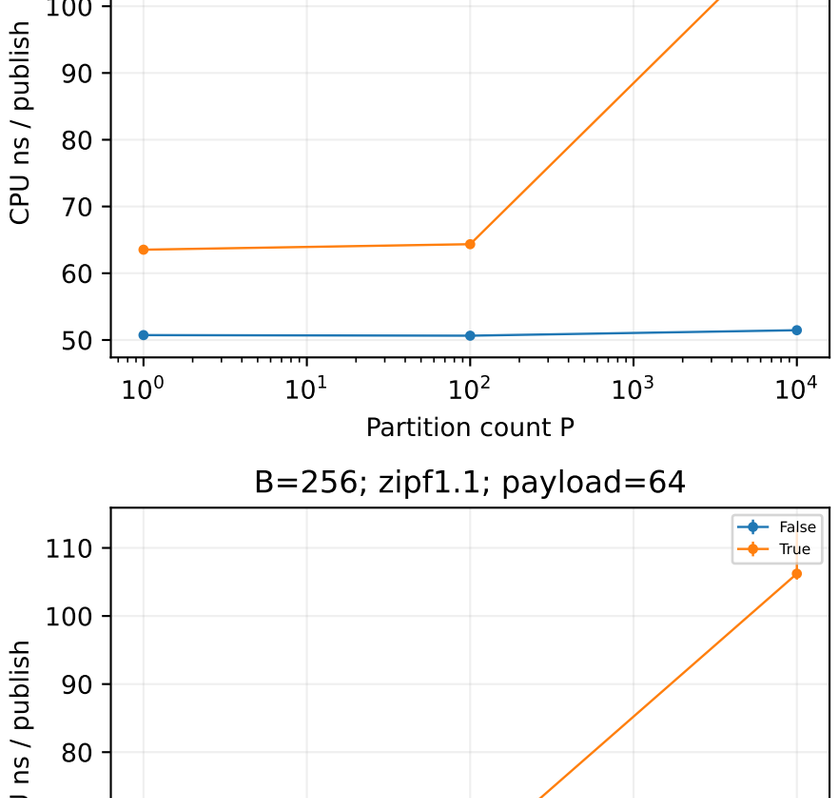


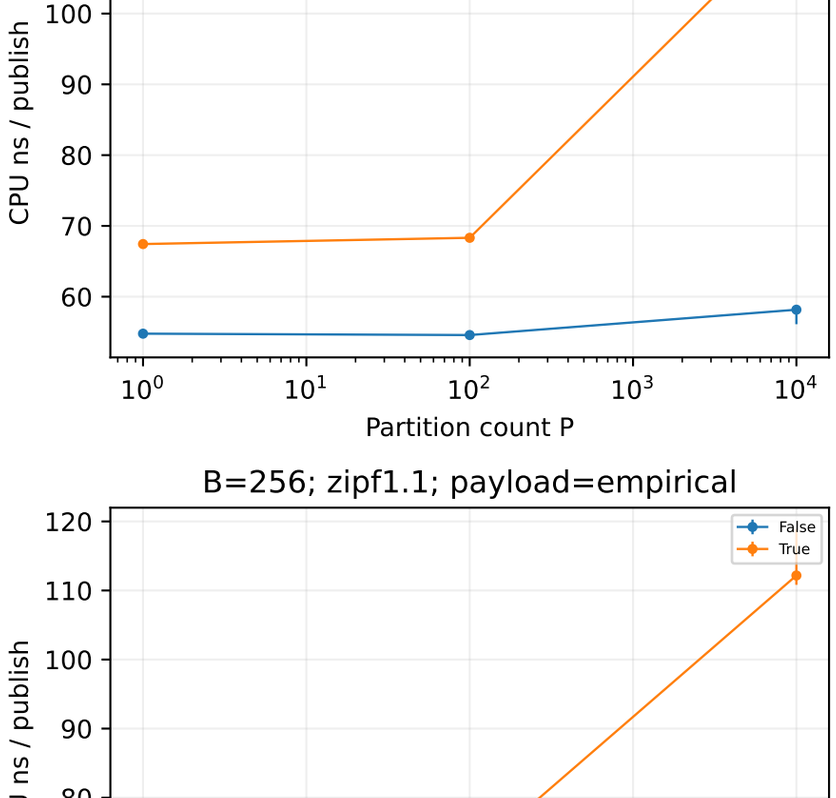


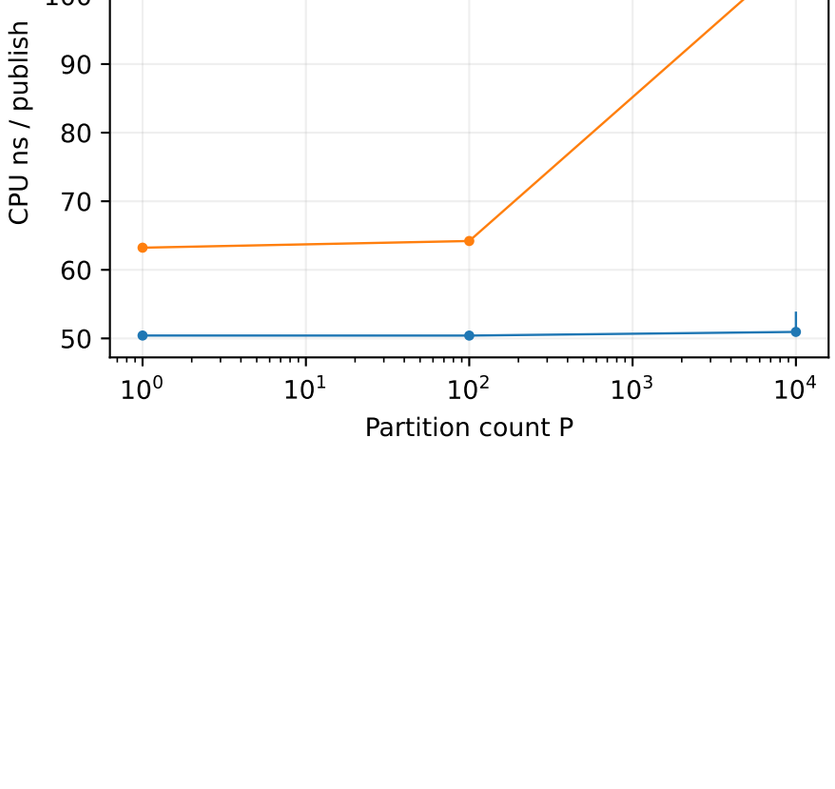


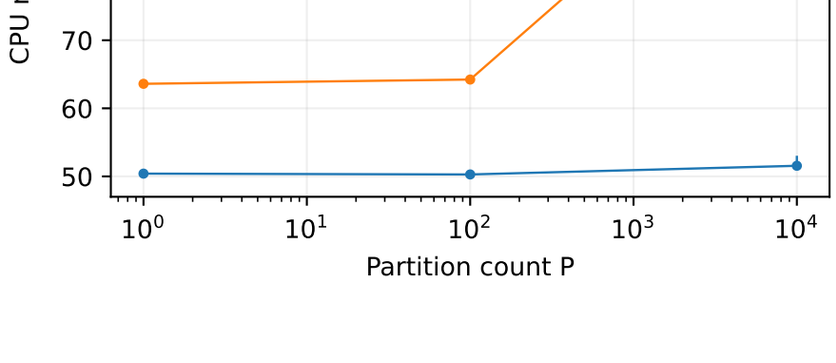


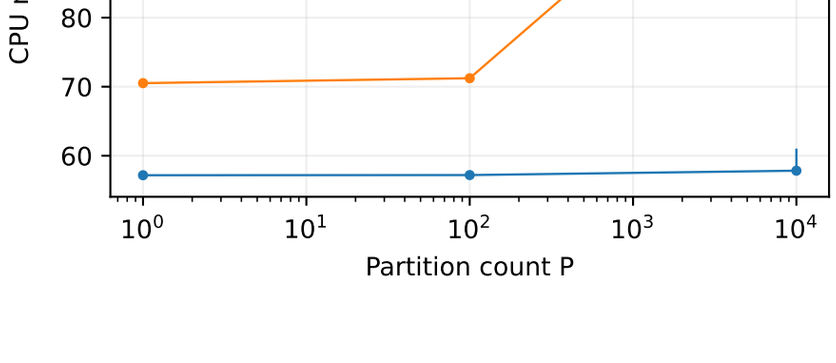












Measured outcomes: ok=1800, non-ok=0; failures retained in tip-summary.json. Error bars: seeded 10,000-resample 95% median bootstrap CIs.